Introduction to Gaussian Processes

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Statistics Background

- Univariate Normal Distribution

$$y \sim \mathcal{N}(\mu, \sigma^2)$$

where μ = Mean and σ^2 = Variance

- Multivariate Normal Distribution

$$\mathbf{Y} \sim \mathcal{MVN}(\boldsymbol{\mu}, \Sigma)$$

where $\boldsymbol{\mu}$ = Mean and Σ = PSD Covariance Matrix

- The marginal and conditional distributions of a MVN are also normal
- Affine transformations of a MVN are also normal

Gaussian Processes

- **Definition:** A Gaussian Process (GP) is a collection of random variables $(f(\boldsymbol{x}))_{\boldsymbol{x} \in \mathcal{X}}$ such that for any finite collection $\boldsymbol{x}_1, ..., \boldsymbol{x}_N \in \mathcal{X}$, the vector $\boldsymbol{f} := [f(\boldsymbol{x}_1), ..., f(\boldsymbol{x}_{\boldsymbol{N}})]^\top$ has a multivariate normal distribution.

$$f \sim \mathcal{GP}(\mu(x), k(x, x'))$$

- $\mu(x)$ is the mean function and $k(x, x')$ is the covariance kernel
- Assume a prior mean function of 0
- GP is typically defined by the choice of Kernel

Gaussian Processes

- Common choices are Matern and Radial Basis Function Kernel; each are parameterized by hyperparameters (e.g., lengthscale ℓ, outputscale η^2, etc.)
- Matern Kernel:

$$k_{Matern}(x, x') = \eta^2 \, \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} d\right)^{\nu} K_{\nu}\left(\sqrt{2\nu} d\right)$$

where $d = (x - x')^{\top}\Theta^{-2}(x - x')$, $\Theta = diag(\ell)$, and D_{ν} is the modified Bessel Function

- RBF Kernel:

$$k_{RBF}(x, x') = \eta^2 \, \exp\left(-\frac{1}{2}(x - x')^{\top}\Theta^{-2}(x - x')\right)$$

Gaussian Processes

Some Kernel Properties:

1. Kernels are symmetric (Covariances are symmetric)

$$K(x, z) = K(z, x)$$

2. K_1, K_2 kernels then $aK_1(x, z) + bK_2(x, z)$ is a kernel for $a, b \in \mathbb{R}^+$
3. K_1, K_2 kernels, then $K_1(x, z)K_2(x, z)$ is also a kernel
4. (2) and (3) imply that any polynomial evaluated at K_1 is a kernel

$$K(x, z) = pol^+[K_1(x, z)]$$

5. (4) implies that $K(x, z) = \exp(K_1(x, z))$ is a kernel

Gaussian Process Regression

- GP Regression is a non-parametric approach to regression problems:
$$y(\boldsymbol{x}) = f(\boldsymbol{x}) \; + \; \epsilon(\boldsymbol{x})$$
- The fundamental assumption is that $f(\boldsymbol{x})$ follows a Gaussian process; place a GP prior on $f(\boldsymbol{x})$
- Provide a training dataset so that the GP model can "learn" and ultimately predict $y(\boldsymbol{x})$
- To obtain GP model that best explains data under prior, optimize the log marginal likelihood using MLE criterion:

$$\log p(\boldsymbol{y} \mid X_{train}) = -\frac{1}{2} y^\top (K_{train} + \Sigma_\epsilon)^{-1} \boldsymbol{y} - \frac{1}{2} \log |K_{train} + \Sigma_\epsilon|$$

Gaussian Process Regression

- To predict at a new test point $x_* \in \mathcal{X}$, compute the posterior:

$$f_*(x_*) := f(x_*) \mid \mathcal{D}_{train} \sim \mathcal{N}(m(x_*), s^2(x_*))$$

with

$$m(x_*) = K(x_*, X_{train})[K_{train} + \Sigma_\epsilon]^{-1}\boldsymbol{y}$$
$$s^2(x_*) = k(x_*, x_*) - K(x_*, X_{train})[K_{train} + \Sigma_\epsilon]^{-1}K(X_{train}, x_*)$$

$$K(x_*, X_{train}) = \begin{bmatrix} k(x_1, x_*) \\ \vdots \\ k(x_n, x_*) \end{bmatrix}^\top$$

Gaussian Process Regression

- Similarly, to predict multiple test points $\{\boldsymbol{x}_{1*}, \ldots, \boldsymbol{x}_{m*}\} = \boldsymbol{X}_* \subset \mathcal{X}$, compute the posterior:

$$f_*(\boldsymbol{X}_*) := f(\boldsymbol{X}_*) \mid \mathcal{D}_{train} \sim \mathcal{MVN}(\boldsymbol{\mu}_*, \Sigma_*)$$

with

$$\boldsymbol{\mu}_* = K(\boldsymbol{X}_*, X_{train})[K_{train} + \Sigma_\epsilon]^{-1}\boldsymbol{y}$$
$$\Sigma_* = k(\boldsymbol{X}_*, \boldsymbol{X}_*) - K(\boldsymbol{X}_*, X_{train})[K_{train} + \Sigma_\epsilon]^{-1}K(X_{train}, \boldsymbol{X}_*)$$

$$K_{(X_{\text{train}}, x_*)} = \begin{bmatrix} k(x_1, x_{1*}) & \cdots & k(x_1, x_{m*}) \\ \vdots & \ddots & \vdots \\ k(x_n, x_{1*}) & \cdots & k(x_n, x_{m*}) \end{bmatrix}^\top$$

Applications of GPs

Bayesian Optimization (BO): A strategy to predict global optimum of black-box functions that are typically expensive to evaluate. Assumptions include the function being continuous and the feasible set to be bounded and low dimension.

- Strategy: Treat unknown objective function as a random function and place a prior over it
- Common choice: Use a **Gaussian Process** as the prior to model function uncertainty
- GPs provide both a predictive mean and a predictive variance

Applications of GPs

BO consists of 2 parts: a Bayesian statistical model and an acquisition function for deciding where to sample next.

A general algorithm is as follows:

1. Apply a GP prior on objective function f with the initial data points n_o in training set X_{train}
2. Assume a simulation budget of N points and some parameter space of potential points H
3. Apply an acquisition function (e.g., Expected Improvement) on H to decide the next point x_n to add onto X_{train}
4. Compute $f(x_n)$ and then refit the GP on f with X_{train}
5. Repeat the process until X_{train} reaches the budget of N

Applications of GPs

One of the most commonly used acquisition functions is **Expected Improvement (EI)**:

- $f(x_n)$ is unknown until computed; ideally want $f(x_n)$ to be "better" than $f_n^* := max_{m \leq n} f(x_m)$
- Find expected value of this improvement and select the next point x_n in order to maximize this improvement
- EI balances exploration (high uncertainty) and exploitation (high mean prediction)

Applications of GPs

Define:

$$EI_n = E_n[[f(x) - f_n^*]^+]$$

This can be evaluated to closed form via Int by Parts resulting in:

$$EI_n(x) = [\Delta_n(x)]^+ + \sigma_n(x)\varphi(Z) - |\Delta_n(x)|\Phi(Z)$$

where $\Delta_n(x) = \mu_n(x) - f_n^*$ and $Z = \frac{\Delta_n(x)}{\sigma_n(x)}$

- φ is the PDF of the Gaussian Dist. and Φ is the CDF of the Gaussian Dist.
- In Step 4 of the BO algorithm, the point with the largest expected improvement x_n is evaluated (ties are broken arbitrarily).

Applications of GPs

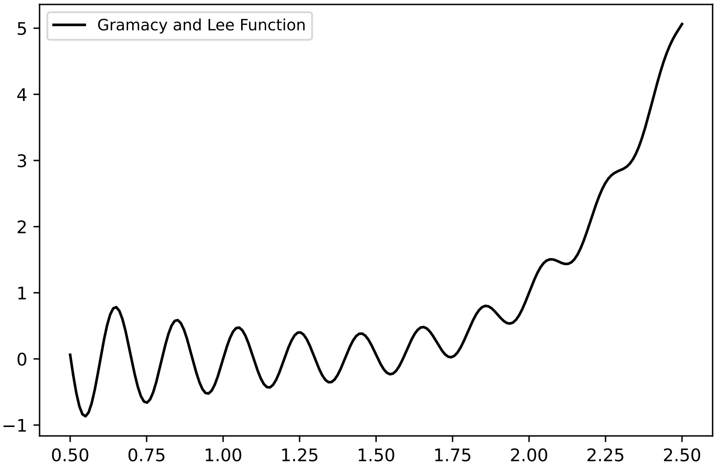

Figure: Gramacy and Lee Test Function: $f(x) = \dfrac{\sin(10\pi x)}{2x} + (x-1)^4$

Applications of GPs

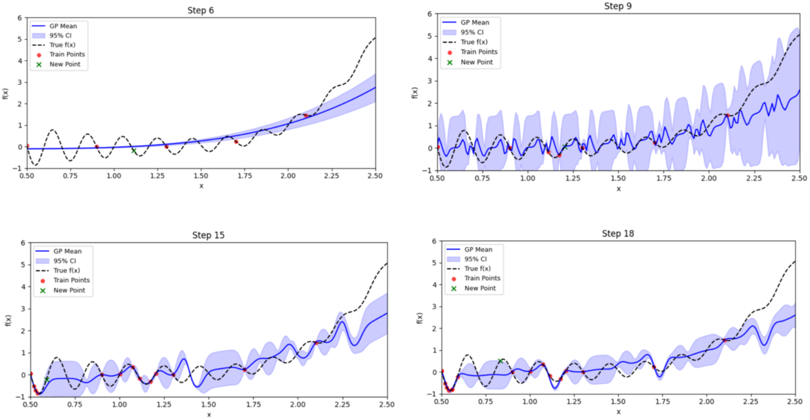

Figure: BO Algorithm: Periodic + Polynomial Kernel and EI

Applications of GPs

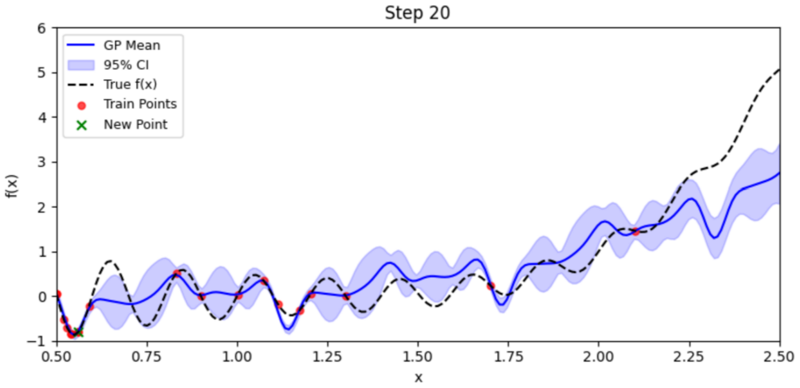

Figure: BO Algorithm: True Argmin at 0.5486 vs BO Argmin at 0.5503